Hack4Insights 2025 – Hackathon Challenge

Project BearCart: E-Commerce Growth & Conversion Intelligence

Industry Context

In today's competitive digital commerce environment, organizations increasingly rely on analytics to understand customer behavior, optimize marketing investments, and improve conversion efficiency. Web traffic data, transaction records, and marketing signals together provide powerful insights when analyzed effectively, enabling businesses to make informed strategic and operational decisions.

Dataset Overview

You are provided with a multi-table e-commerce dataset representing an online retail platform selling consumer products through its website. The dataset captures website sessions, user interactions, marketing traffic sources, product details, orders, and refunds. The data reflects real-world business conditions and includes **data quality issues** that require cleaning and preparation before analysis.

Business Scenario

BearCart, a growing online retail company, is preparing for its next phase of expansion. While the business collects large volumes of digital and transactional data, leadership lacks clear visibility into customer behavior, marketing effectiveness, and revenue performance. Your team has been brought in as data analytics consulting unit to clean the data, analyze performance, and present insights that can support better business decision-making.

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Case Study Objectives

Participants are expected to:

- Prepare and clean raw datasets for analytical use
- Analyze website, marketing, and sales performance
- Identify key drivers of conversion and revenue
- Translate data insights into meaningful business outcomes
- Communicate findings effectively through dashboards

Hackathon Challenges (Problem Overview)

Teams are expected to explore and address the following areas:

1. Data cleaning and preprocessing of raw datasets
2. Analysis of website traffic and order trends
3. Evaluation of session-to-order conversion behavior
4. Comparison of marketing channel performance
5. Assessment of revenue and order value trends
6. Analysis of refunds and their business implications
7. Development of actionable, data-backed business insights

Dashboard Requirement (Overview)

Each team must design an **interactive dashboard** that communicates insights clearly to business stakeholders. The dashboard structure, visual design, and storytelling approach are left to the team's discretion. Participants may use any tools or platforms they are comfortable with.

Indicative KPI Coverage

Dashboards should broadly reflect:

- Website engagement and traffic metrics
- Conversion performance indicators
- Revenue and order value trends

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- Marketing channel effectiveness
- Product-level performance insights

Evaluation Criteria

Teams will be evaluated based on the following high-level parameters:

- **Data Preparation & Cleaning** – Quality of data handling and preprocessing
- **Analytical Depth** – Ability to derive meaningful and accurate insights
- **Business Relevance** – Practicality and clarity of insights and recommendations
- **Dashboard Effectiveness** – Visual clarity, usability, and storytelling
- **Innovation & Approach** – Creativity in analysis, metrics, or presentation
- **Overall**
- **Clarity** – Communication of findings and conclusions

Expected Deliverables

Each team must submit:

- Cleaned and processed datasets
- An interactive analytical dashboard
- An insight summary
- Final business recommendations

Reflection Points

- How did data quality affect analytical outcomes?
- Which metrics were most influential in decision-making?
- What assumptions were made during analysis?
- What additional data could further improve insights?

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